

TrES-3b

R-1888

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_140527 · created 2026-08-03T12:38:14+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456804.8164 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.025
Transit depth (Rp/Rs)^2	3.85 +/- 1.0 [%]
Orbital Inclination (inc)	82.33 +/- 0.89
Airmass coefficient 1 (a1)	0.841 +/- 0.021
Airmass coefficient 2 (a2)	0.166 +/- 0.044
Scatter in the residuals of the lightcurve fit is	2.44 %
Best Comparison Star	#1 - [428, 416]
Optimal Method	PSF photometry
Transit Duration (day)	0.0621 +/- 0.0077

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.196 ± 0.025 vs 0.16309 (deviation 1.18σ)
Duration fit vs published	0.0621 d vs 0.0567 d (deviation 0.63σ)
Mid-time O-C	1.8 min
Depth SNR	7.0
Residual scatter	2.44 %

Light curve

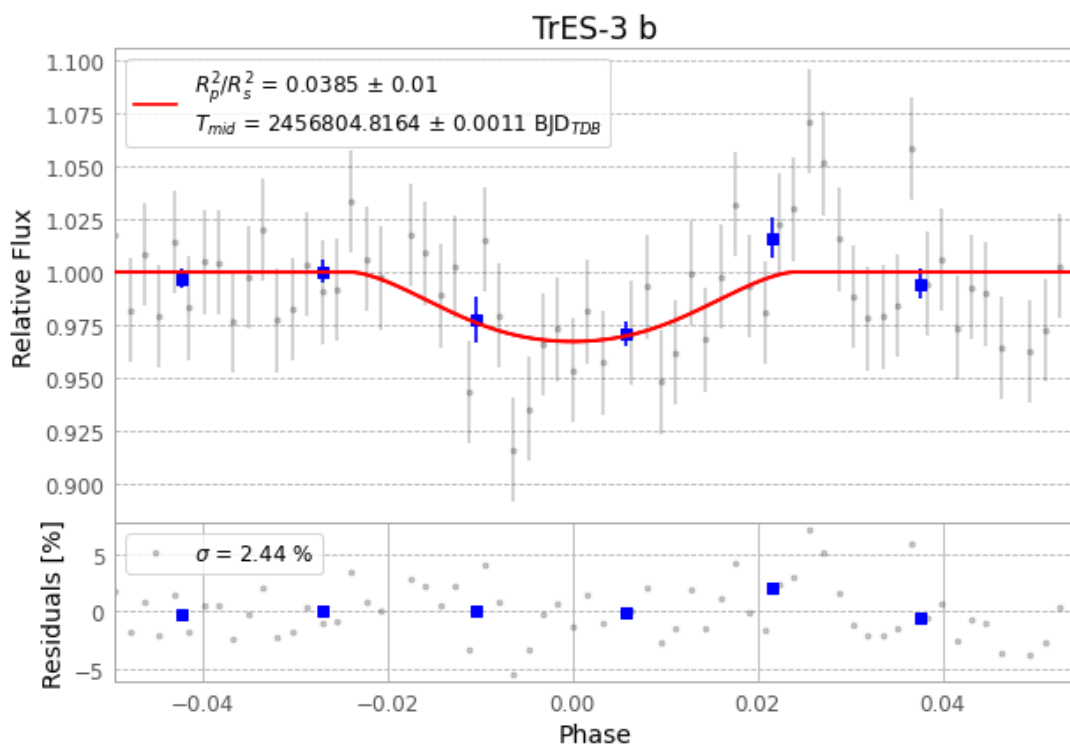


Figure 1 — FinalLightCurve_TrES-3 b_2014-05-26.png (SHA-256 8d56d8c7f5f8917b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [322, 261], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 083ebb6230f6f90e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

66 FITS frames (43 MB), TRES-3140527055712.FITS → TRES-3140527091212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-140527.log (1456614f0140...), FinalLightCurve_TrES-3 b_2014-05-26.png (8d56d8c7f5f8...), FinalLightCurve_TrES-3 b_2014-05-26.csv (fb0f1afdafd2...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 12:38Z.